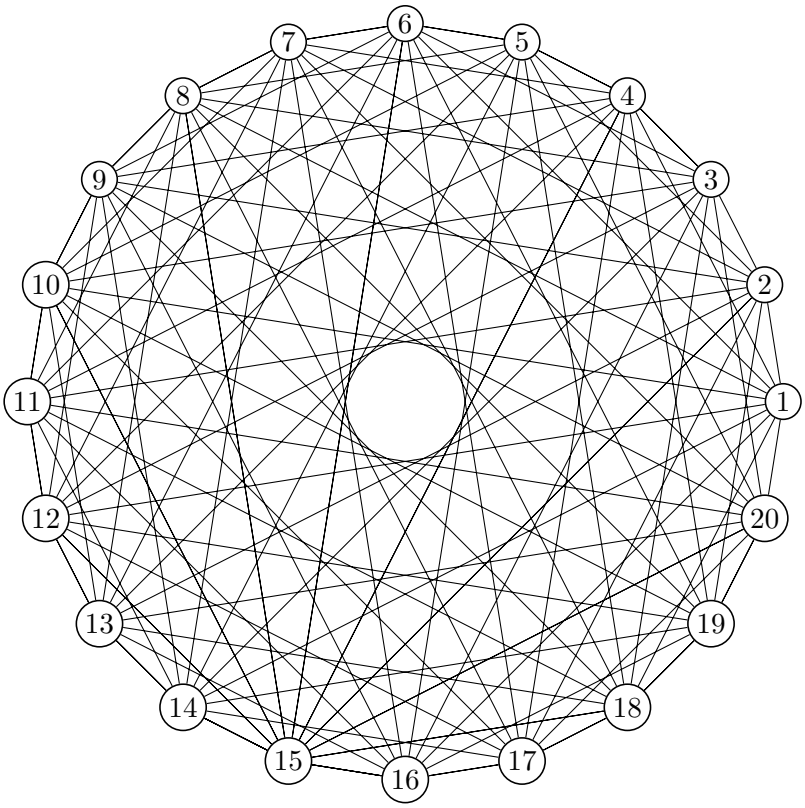
[B.A.S.H] **October 15, 2026** **Clarifier:** Henry
Manager: Seungwon
Reporter: Rachel & Aadya
Skeptic: Benjamin

Report: Today, felt pretty relaxed as we got many problems done and worked together. Number 1 and 2 were quick, in general, though they required some discussion/time to make the graphs. 3 and 4 were also done fast since many of the ideas were similar to 1/2 and to examples in-class. For number 5, we were able to illustrate our example from in-class, but we did not have enough time to prove each of the conditions. 6 was also easier, so we finished that alongside 7 hehehehehe and part of question 8 (still have to prove the necessary condition). Number 9 was also simple to write-up, so we had time to create a graph for 10. Finally, we completed 12 and 15c. Thank you to everyone who made this program possible!
NOW ONTO THE MORE IMPORTANT RACHEL ADDITION:
So like someone in this group (I am not a snitch) gave me some chewing gum so that I may be able to write their report for them. The person in the group was busy hanging out with their friend Parabola who they met at a function once. Anyways, our group worked with some other people from another group (Spies I fear) to solve our problems and then we went back to our respective groups for the write up process. It was a pretty chill day. I mean we got side tracked a little with doing the dotted line thingy-thing on the chalkboards but who wouldn't when we got like 10 new sticks of chalk! Anyways, we all really loved the course (hance why I am here four years later still being a reporter) and thank you all for being here for us these past two weeks!! I never thought that this part of the homework would get all meta. (Original meaning of the word, not the company). But there we are. I paid the money, I take the ride.

SOLUTIONS & PROOFS:
Without Loss of Generality is abbreviated as WLoG within this document.

1. $n = 20$, Brualdi graphs where a, b is an edge IFF $a + b$ is odd. Great work!



[Graph 1]

(a) Is this graph bipartite?

Solution. Since even vertices are connected odd vertices and not other even vertices, this graph is bipartite as the even vertices (ex: 2, 4, 6...) are in one set, and the odd vertices (ex: 1, 3, 5...) are in another set.

(b) Is it connected?

Solution. Yes, to get from an even vertex to an odd, there is a shared edge, and to get from an even to even or odd to odd, there are two edges between even and odd that can be taken one after each other.

(c) Are all components Eulerian?

Solution. Since Eulerian is connected and even-degreed, we look to satisfy both these conditions. As shown above, the graph is connected, and because each vertex has degree 10 (when added 10 * 20 = 200), the graph is even-degreed as well. Therefore, the graph is Eulerian.

(d) Are all componenets Hamiltonian?

Solution. Yes, this graph is hamiltonian because you can alternate by passing through even and odd vertices through established edges, with each move being unique.

(e) Compute $\chi(G)$.

Solution. Since the even vertices are not connected to each other and neither are the odd vertices, they form two independent groups, meaning there are two colors needed.

(f) Compute $\alpha(G)$.

Solution. Each independent set has 10 vertices (10 odd vertices and 10 even vertices), therefore, $\alpha(G)$ is 10.

(g) Compute $dom(G)$.

Solution. The domination number is 2 because one odd vertex and one even vertex can be chosen.

(h) Compute $\omega(G)$.

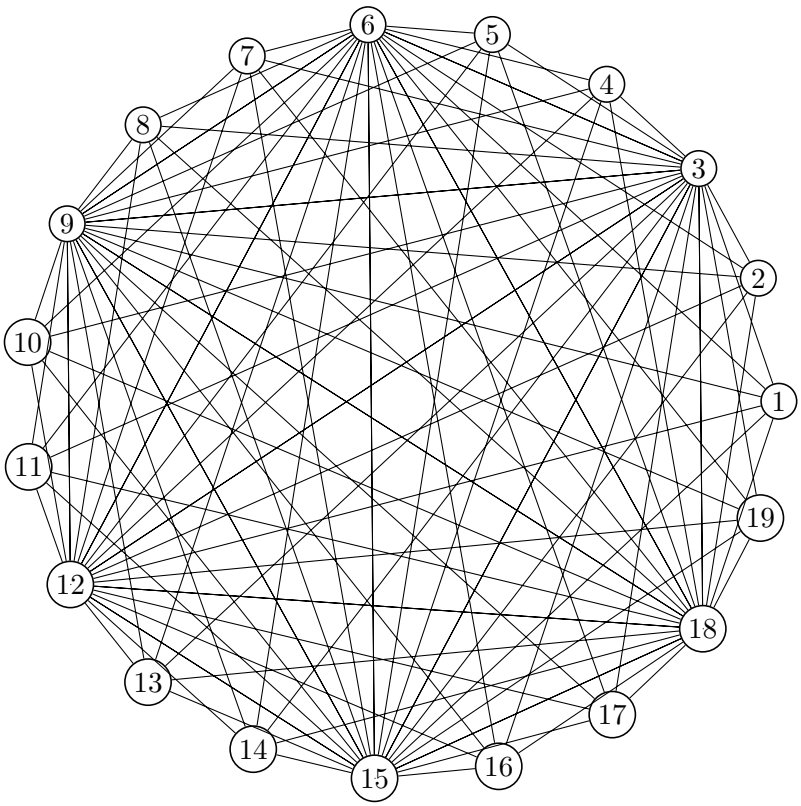
Solution. The clique number is 2 as well because there are no complete graphs above K_2.

(i) Compute $\theta(G)$.

Solution. The clique partition number is 10 since $20/2 = 10$.

RW: Very nice work. I like your graph (and the one below!)

2. $n = 19$ Brualdi graphs where a, b is an edge IFF ab is a multiple of 3.



[Graph 2]

(a) Is this graph bipartite?

Solution. This graph is not bipartite because it would have to split into more than 2 independent sets. I would have actually accepted "no" here, but your explanation is circular enough that I have to ask: why can't it?Do you see how that was a circular argument?

(b) Is it connected?

Solution. Yes, the graph is connected. From any vertex, you can pass through a vertex of multiple three and get to any other vertex.

(c) Are all components Eulerian?

Solution. Yes, the graph is connected and is even degreed because the vertices of multiple three have degree 18, and the other vertices have degree of 6 (meaning the graph, overall, is even-degreed).

(d) Are all components Hamiltonian?

Solution. No, the graph is not hamiltonian because there are not enough vertices of multiple three to get to all of the remaining 13 vertices without double-passing. Almost sounds like toughness...

(e) Compute $\chi(G)$.

Solution. The chromatic number is 7 because there is a color assigned to each of the vertices of multiple three and one other color for all of the vertices that are not multiples of three.

(f) Compute $\alpha(G)$.

Solution. The independence number is 13 because there are 13 vertices (19 - 6) in the ind. set of vertices that are not multiples of three.

(g) Compute $dom(G)$

Solution. The domination number is one because choosing any vertex of multiple three will connect to all other vertices.

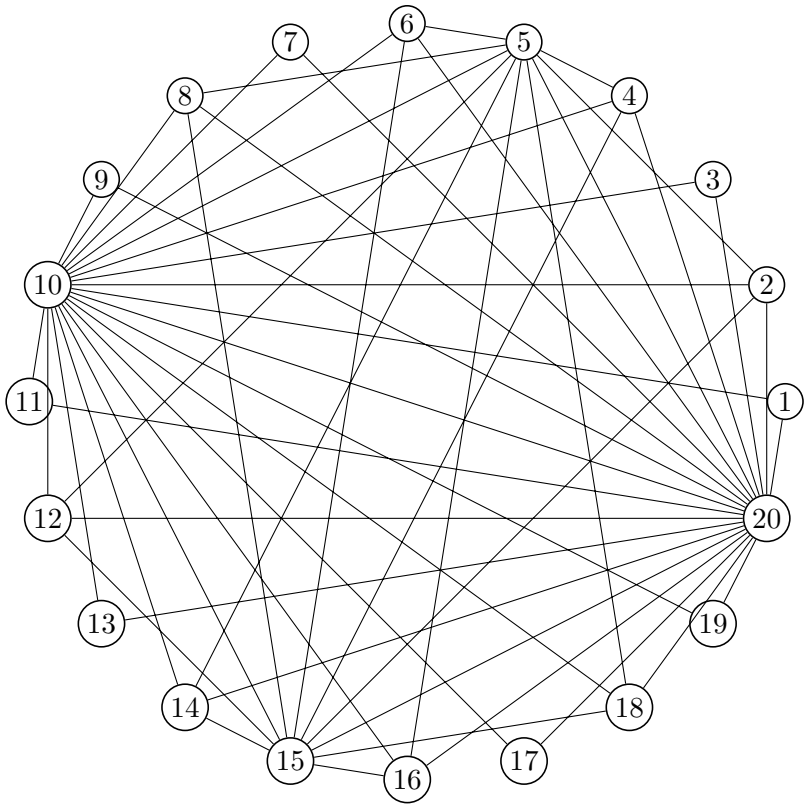
(h) Compute $\omega(G)$.

Solution. The clique number is 7. It contains 3, 6, 9, 12, 15, 18 and another vertex that is not a multiple of three.

(i) Compute $\theta(G)$.

Solution. The clique partition number is 13 because everything can be removed besides the K_7.

3. $n = 20$ Brualdi graphs where a, b is an edge IFF ab is a multiple of 10.



[Graph 3]

(a) Is this graph bipartite?

Solution. No. There is an odd cycle, 8-10-20-8

(b) Is it connected?

Solution. Yes. 20 is connected to everyone.

(c) Are all components Eulerian?

Solution. No. 10 and 20 have odd degrees.

(d) Are all components Hamiltonian?

Solution. No. every odd number that is not a multiple of 5 has 2 degree, and they are only connected to 10 and 20. Almost sounds like toughness...RW: I am getting deja-vu

(e) Compute $\chi(G)$.

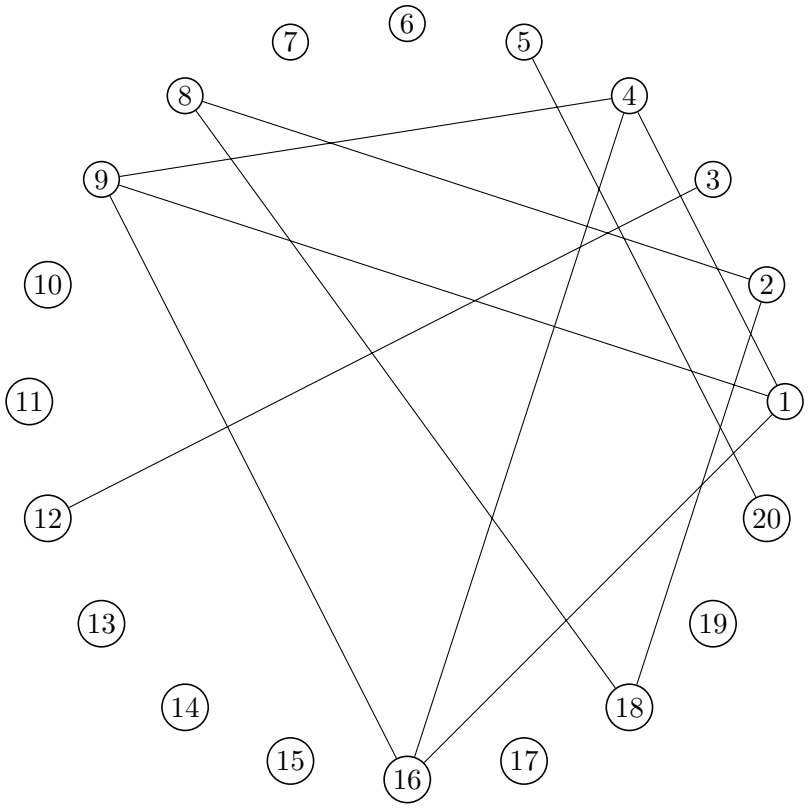
(f) Compute $\alpha(G)$.

(g) Compute $dom(G)$.

(h) Compute $\omega(G)$.

(i) Compute $\theta(G)$.

4. $n = 20$ Brualdi graphs where a, b is an edge IFF ab is a perfect square.



[Graph 4]

I obviously love your graphs. But is the figure above the best illustration of a graph consisting of a K4, some K3s, some K2s, some isolated vertex? If you do an illustration, pretty is great, but the most important thing is clarity for the reader. When I draw this in class today, my graph will not be as gorgeous, but it will be clearer as to what is going on.

(a) Is this graph bipartite?

Solution. Since this graph has K_4, this graph is not bipartite.

(b) Is it connected?

Solution. No. This graph is consisted of K_4, K_3, and two P_2.

(c) Are all components Eulerian?

Solution. It is all Eulerian except for K_4. P_2 is not.

(d) Are all components Hamiltonian? P_2 is not.

Solution. Yes.

(e) Compute $\chi(G)$.

Solution. 3 There is a K_4 so uh...?

(f) Compute $\alpha(G)$.

Solution. 11

(g) Compute $dom(G)$.

Solution. 11

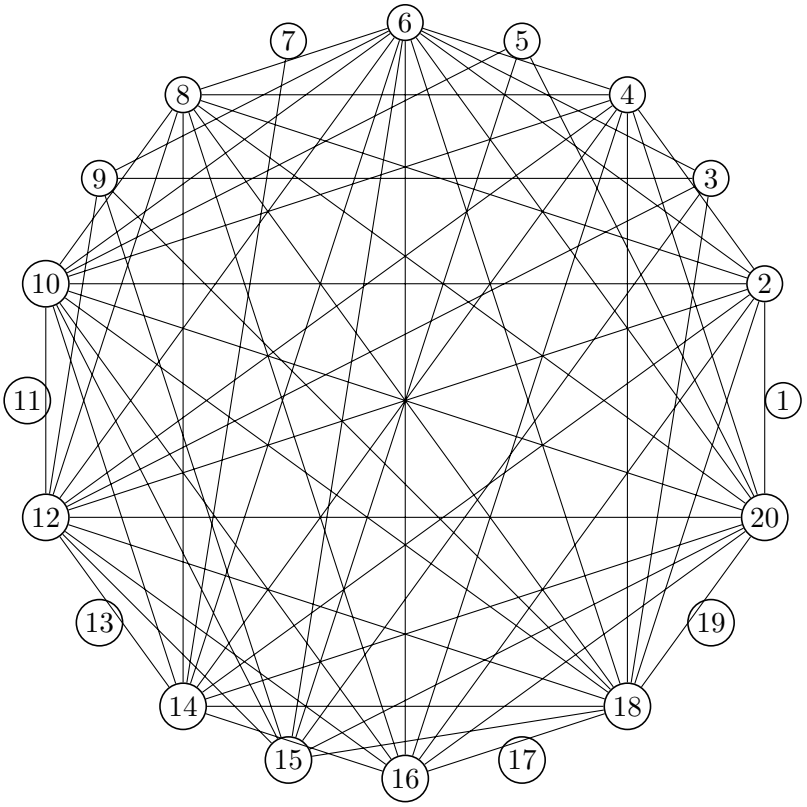
(h) Compute $\omega(G)$.

Solution. 4

(i) Compute $\theta(G)$.

Solution. 11

5. $n = 20$ Brualdi graphs where $\gcd(a, b) \neq 1$. Cool graph!



[Graph 5]

(a) Is this graph bipartite?

(b) Is it connected?

(c) Are all components Eulerian?

(d) Are all components Hamiltonian?

(e) Compute $\chi(G)$.

(f) Compute $\alpha(G)$.

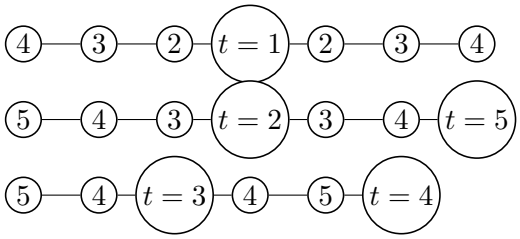
(g) Compute $dom(G)$.

(h) Compute $\omega(G)$.

(i) Compute $\theta(G)$.

6. What is the burning number of $P_7 + P_7 + P_6$?

   Solution. .



[Graph 6]

It's 5. Why can't it be 4?

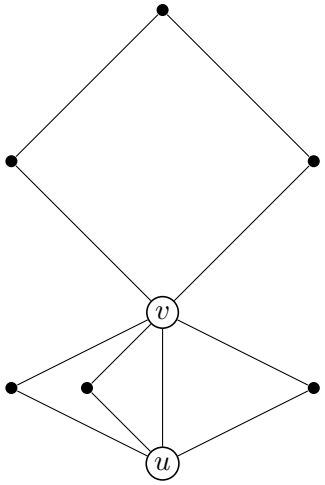
7. Prove that the cop number of a tree is 1.

   Proof. Start from any vertex. Because there is only 1 path to robber, move to the vertex that is to robber. The distance between robber and cop will be still same if robber has a vertex that helps robber to run away. But, that won't be always true. Because tree has leaves, robber can't run away from the cop and the distance will decrease. At the end, cop can catch robber if cop always go to robber. Good! □

   RW: Nice and thorough.

8. A vertex u is called a corner if there is a vertex v such that every vertex adjacent to u is also adjacent to v. Is having a corner a necessary condition for a graph to be a cop-win graph? Is it sufficient? As always, prove your answer.

   Proof. We can show that having a corner is not a sufficient condition in a cop-win graph through the graph below: Why does the robber win here? Just a couple words will do.



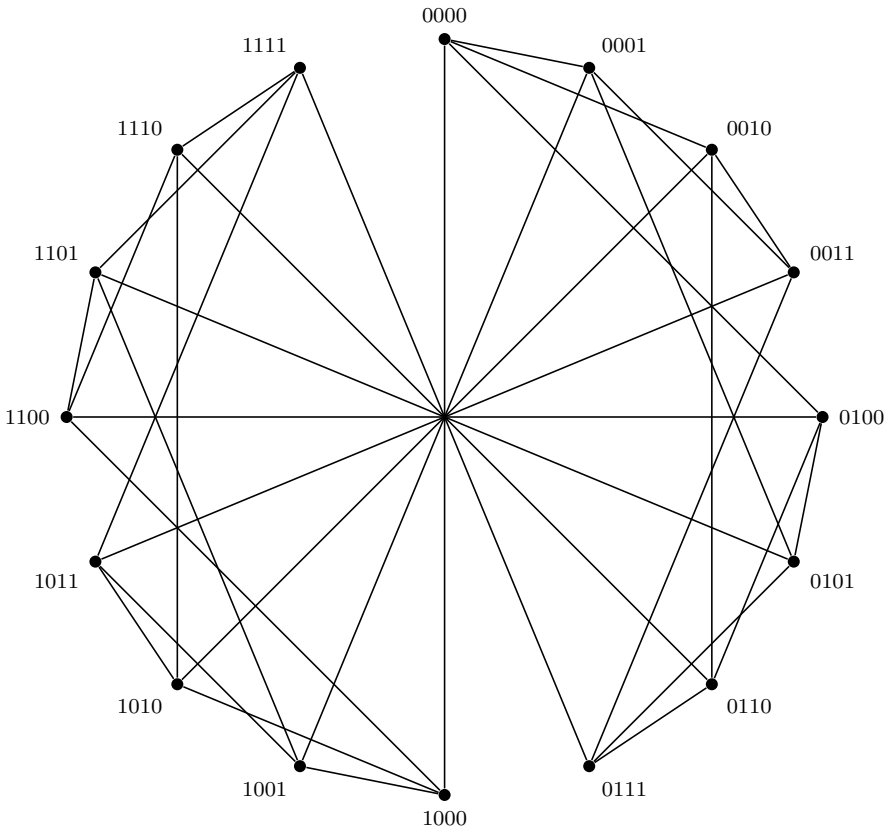
[Graph 7]

□

9. What is the relationship between the domination number of a graph and its cop number?

   Solution. The domination number shows the minimum number of cops needed in order to definitely catch the robber within one move.

10. Q_4 is the hypercube.



[Graph 8]

   (a) Show the cop number is ≤ 4.
   (b) Show the cop number is > 2.
   (c) Find the cop number of Q_4.

11. Is there a k-regular graph with a cop number greater than k?

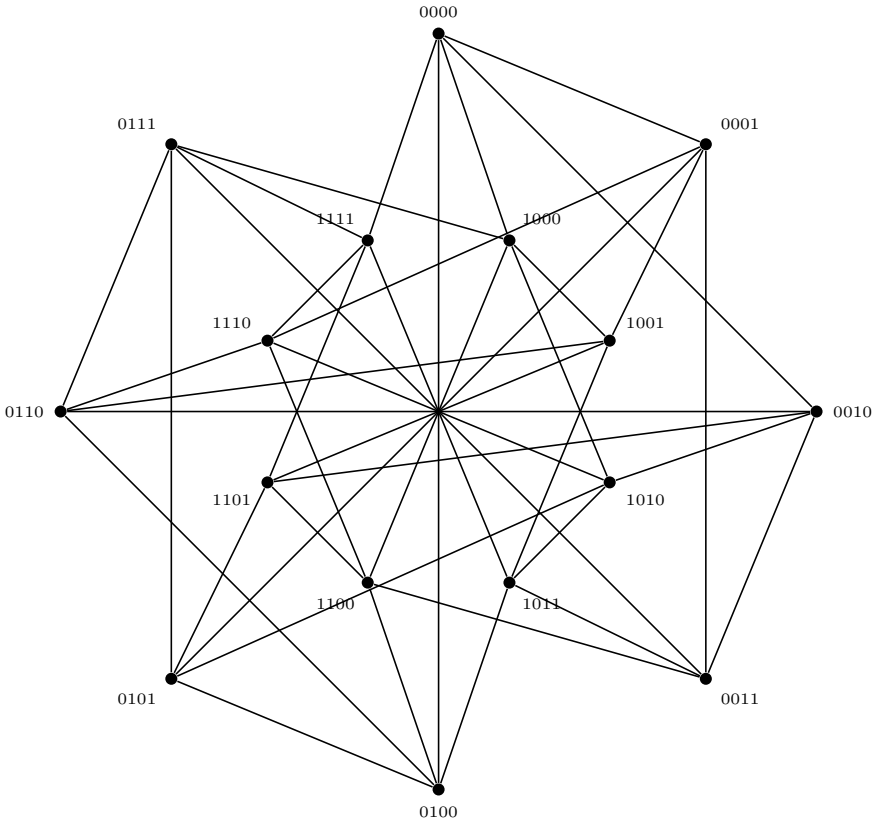
12. Do all chordal graphs have cop number 1?

    Solution. Every chordal graph has cop number 1. This is because chordal graphs have a special structure that allows a cop to always move closer to the robber and eventually catch them. In a chordal graph, there is always a way to remove vertices step by step while keeping the same capture ability, meaning the graph can be simplified without making it harder for the cop to win. Therefore, one cop is enough to catch the robber on any chordal graph, so the cop number of every chordal graph is 1. How does the cop do this? It's the right idea, but needs explanation.

13. Is there a bound on the cop number of a planar graph? If so, what is it? You may not have an airtight proof but give your reasoning.

14. Here is something professor Shaw's been looking into. What can you figure out about cop numbers of large cycle graphs with k chords? If $k = 0$, the cop number is 2. That's just a cycle. If $k = 1$, the cop number is also 2. You can place one cop at each end of the chord. I have a proof that if $k = 2$ the cop number is also 2. What do you think about larger values of k?

15. Below is a Clebsch graph.

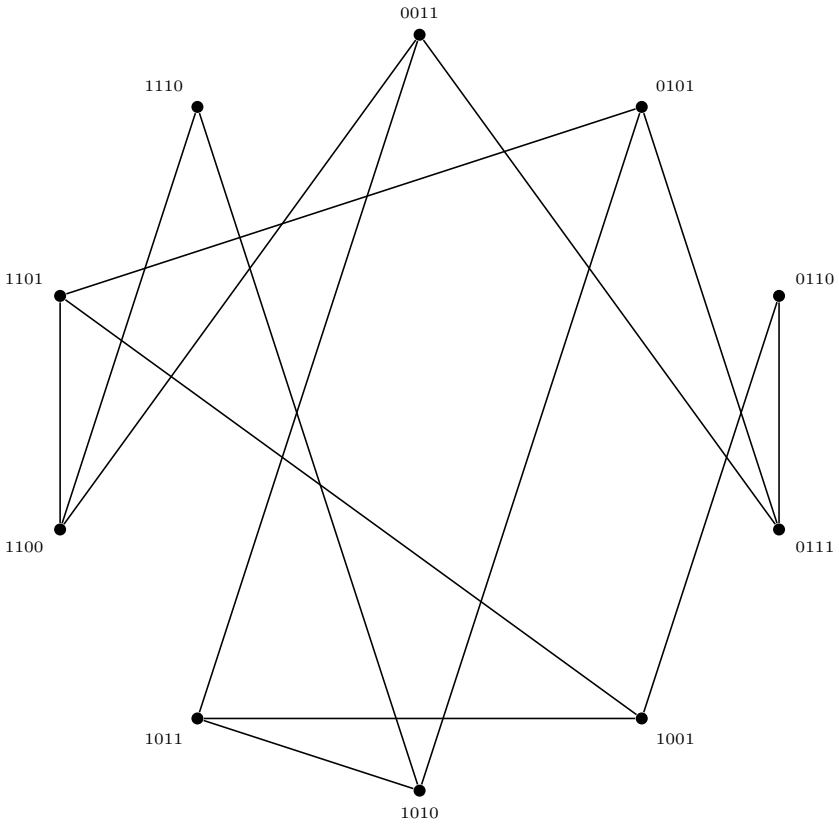


[Graph 9]

(a) Find our five invariants for the Clebsch grpah.

(b) What is the minimum number of vertices you need to remove in order to disconnect it?

(c) The graph below is the graph made by removing one vertex and its direct neighbors from the clebsch graph. There is something interesting to say about it, can you find it?



[Graph 10]

16. For each member of your group : What is your favorite quotation and why?



Figure 1: Guess who?

Solution. Benjamin : "Six feet tall and ruggedly handsome" - Prof. Doug Shaw, sometime in the 21st century. I probably don't require a reason for this. Early, early, 21st centuryRW: sometimes facts speak for themselves.

Aadya: "RAWR" - Prof. Shaw...I look forward to witnessing this again one day
RW: Aadya, please do not encourage this LOL. That was single-handedly the most weird thing I have ever seen this man do, and well I've known him for only several years, but I still think that speaks volumes after hearing the same 11 lectures five times! (Gah sorry about the run on sentence there!) I have no memory of what this refers to.

ANYWAYS! It was a real pleasure being your undergrad TA you all! I really think you guys put forth some **very** high quality work that I hope you are proud of. Whatever comes next for you all, please stay curious and I wish you the best! - Rachel

I really liked working with you all. I loved how you rolled with me whether I was doing hard math, easy math, or just rambling on. Thank you for teaching me to do things with Overleaf that I've never done. I hope you all have a great fall!